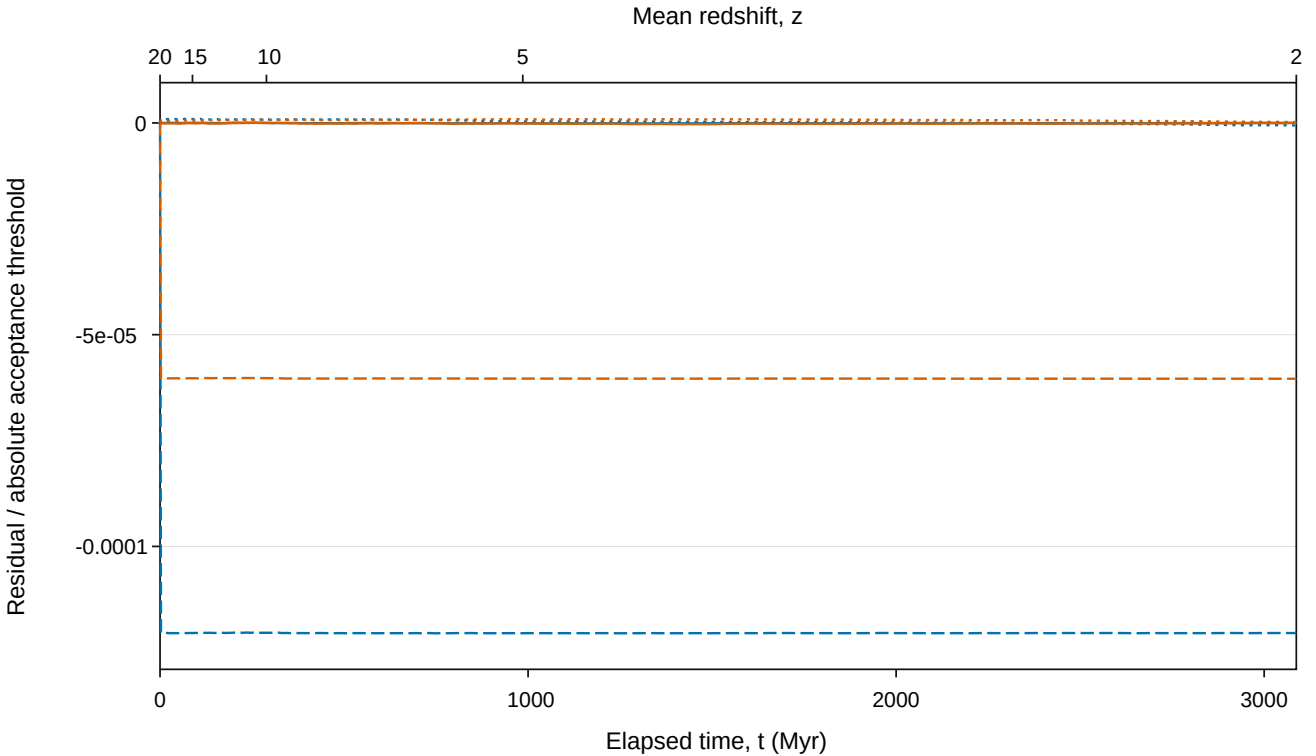
(j) Signed open-system conservation residuals



Baryon + external ledger; heavy charge - electron ledger; initial values removed.
Original absolute threshold = 1.0820350005000101e-8 particles / Href.
Source: saved pilot17_populations.csv and three-case NPZ arrays.